

Assignment - 1

Program - 1 - Design and implementation of a FULL ADDER using VHDL

⇒ Inputs : A, B, Cin

⇒ Outputs : Sum (S), Cout (Carry out)

⇒ Truth table :

A	B	Cin	Sum (S)	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

⇒ Program :

Library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity full_adder is

Port (

A : in STD_LOGIC; -- Input A

B : in STD_LOGIC; -- Input B

Cin : in STD_LOGIC; -- Carry input

Sum : out STD_LOGIC; -- Sum output

Cout : out STD_LOGIC -- Carry output

);

end full_adder;

architecture Behavioral of full_adder is

begin

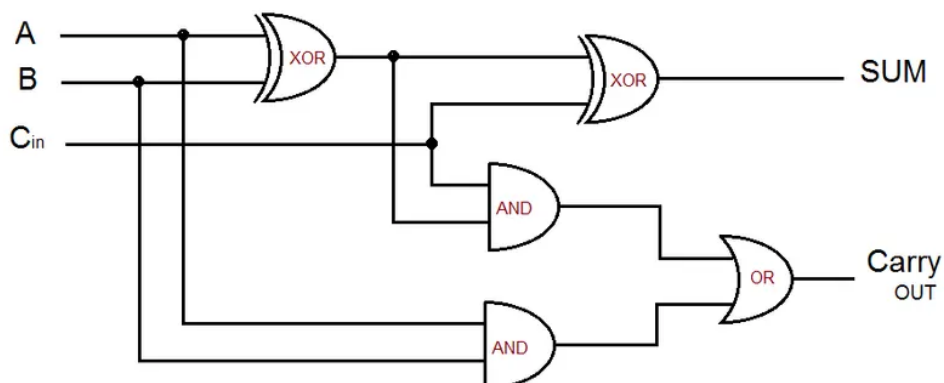
Sum <= A XOR B XOR Cin;

-- Sum = $A \oplus B \oplus Cin$

Cout <= (A AND B) OR (Cin AND (A XOR B));

-- Cout = $AB + Cin(A \oplus B)$

end Behavioral;

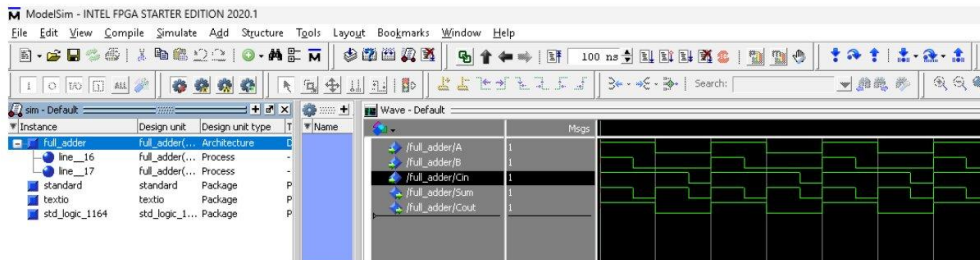


Full Adder Simulation Results

1. Full Adder Simulation using Clock Inputs

This simulation uses periodic clock-driven inputs with different duty cycles:

- A: 50% duty cycle clock
- B: 30% duty cycle clock
- Cin: 80% duty cycle clock



2. Full Adder Simulation using Forced Input Values

This simulation verifies the full adder by manually forcing input values from the ModelSim Transcript using commands.

